DS4300 HW3 Report: Elliot Jerng and Ava Knight

In order to teach other data scientists about Mongo and its storage and query capabilities, we decided to create an interactive dashboard showing COVID-19 trends across the world over different parameters. Data scientists analyzing the data will not need to know Mongo or how to query in order to glean information and make conclusions about the data we're presenting, and we hoped to make analyses about this data accessible to those without a data or programming background as well with clear visualizations and basic analyses attached. Users do not need to call any Mongo queries themselves, but can still play around with different parameter selections to analyze broader trends.

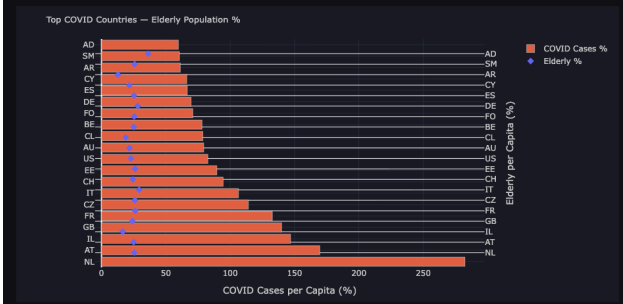
Looking at the data available, there were many interesting factors that could be analyzed for their effect on covid-19 cases. In particular, factors such as elderly population, urban population, population density, and human development index (HDI) were of particular interest. We chose these factors because we believed they had the most potential for being impactful on covid-19 cases, and would reveal interesting trends that connected socioeconomic and epidemiology factors. Looking at the elderly population, we can look at the countries with the most covid cases per capita, and look at their elderly populations per capita. We also looked at the opposite, so look at what countries have the most elderly per capita, and look at their covid cases per capita. Looking at urban and non-urban populations, we can see what are the most urban populations, and what are their covid cases per capita. Additionally, we can see what are the least urban populations, and what are their covid cases per capita. Looking at the most and least dense populations we can analyze the most densely populated countries, and their covid cases per capita. Additionally, we analyzed the least densely populated countries, and their covid cases per capita. Looking at wealthiest vs poorest (with HDI) countries, we chose to ask, what are the highest HDI countries, and what are their covid cases per capita? And what are the lowest HDI countries, and what are their covid cases per capita? These two questions can also be analyzed from the point of view of death rates. The aggregations of queries was the result of the implementation of various techniques, including aggregation, filtering, and sorting.

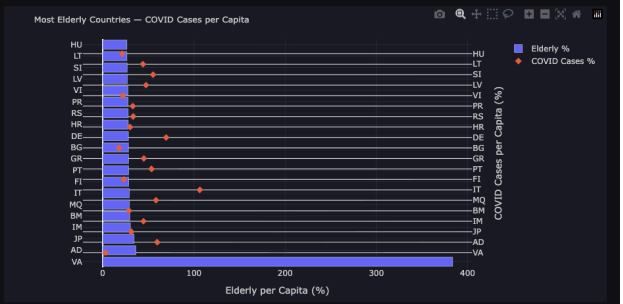
<u>Contributions:</u>

Elliot wrote the queries and Ava created the dashboard connection and layout. We both analyzed the data from both the query results and the resulting visualizations to come to some basic conclusions about overall trends, and also assisted each other with initial conceptualization, programming, and debugging.

<u>Visualizations:</u>

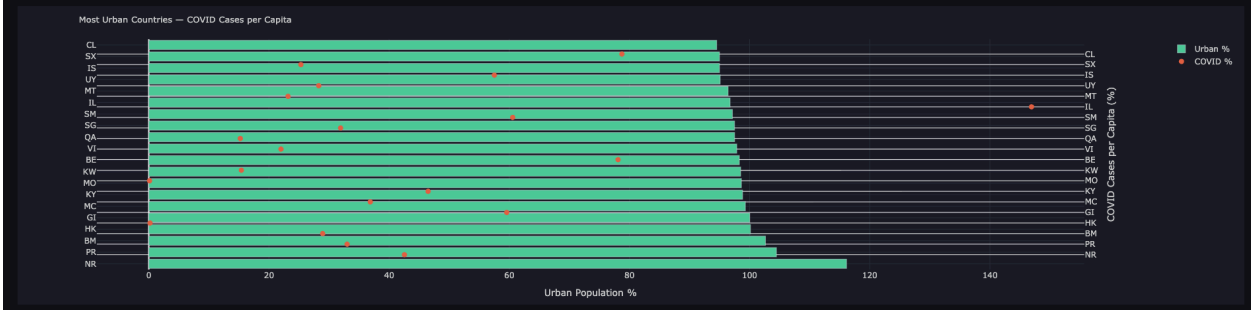
Elderly Population Analysis





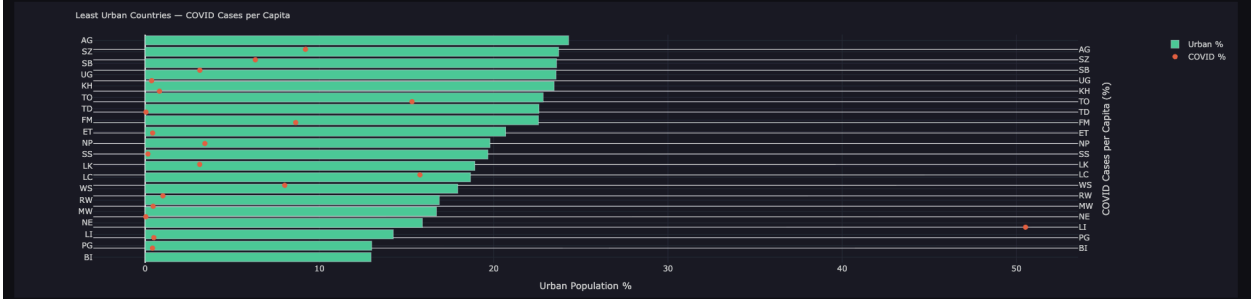
Urban vs Non-Urban Population

Most Urban Least Urban



Urban vs Non-Urban Population

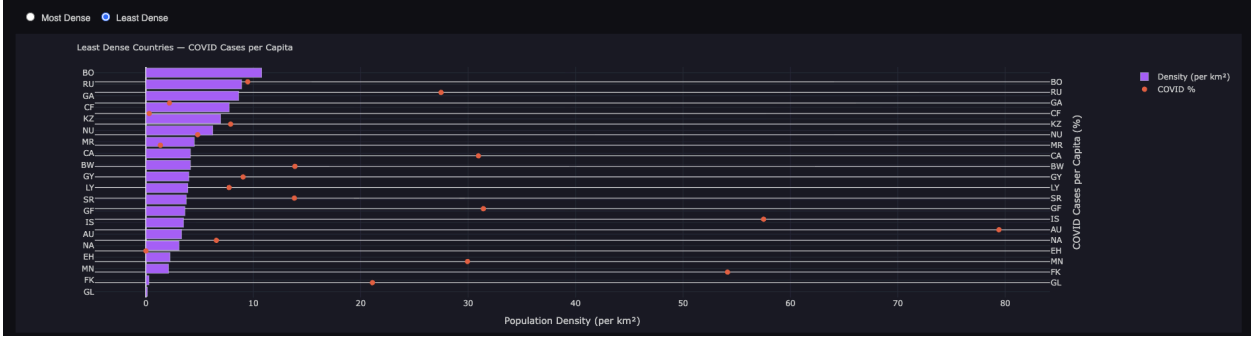
Most Urban Least Urban



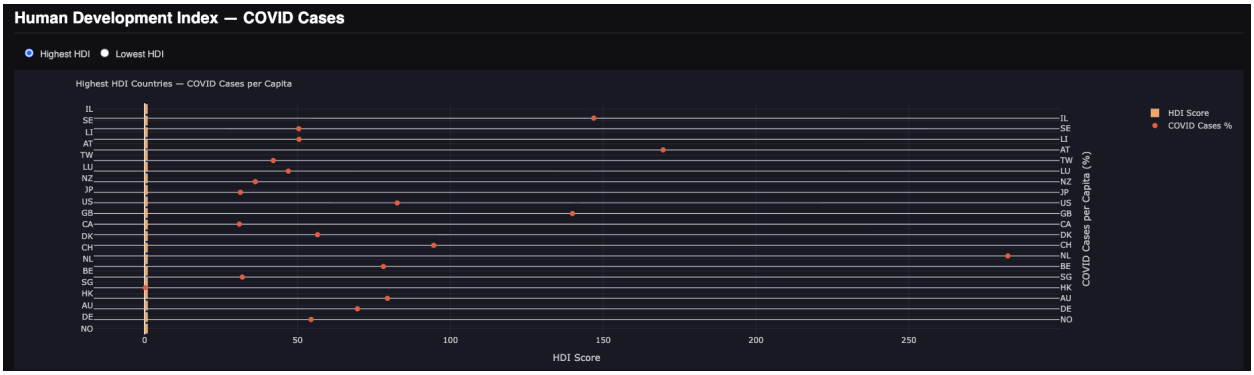
Population Density

Most Dense Least Dense

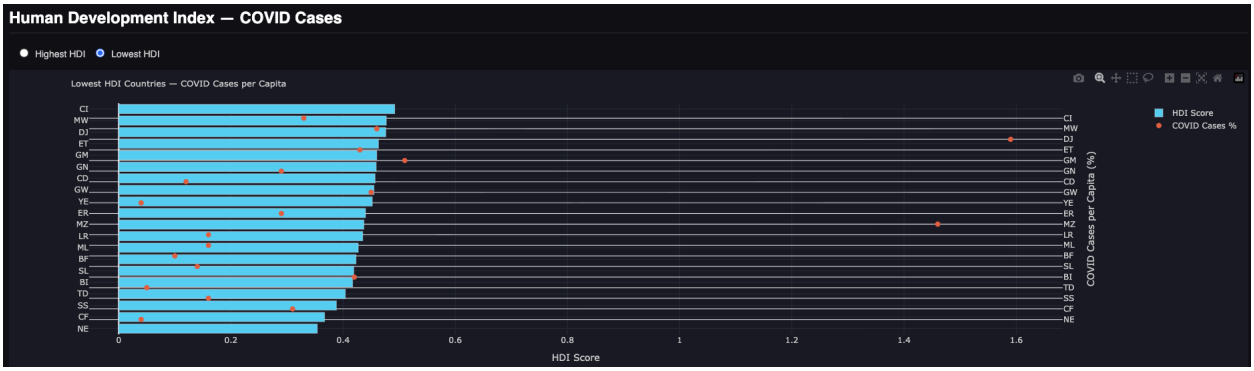
Population Density



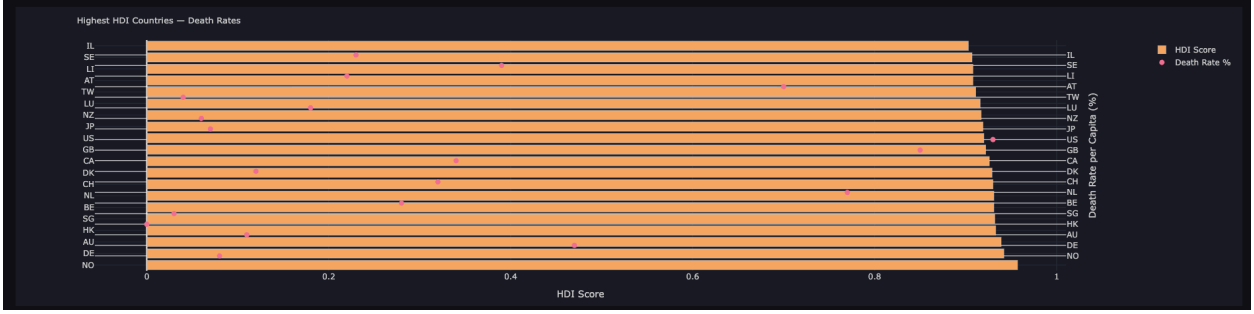
Human Development Index — COVID Cases

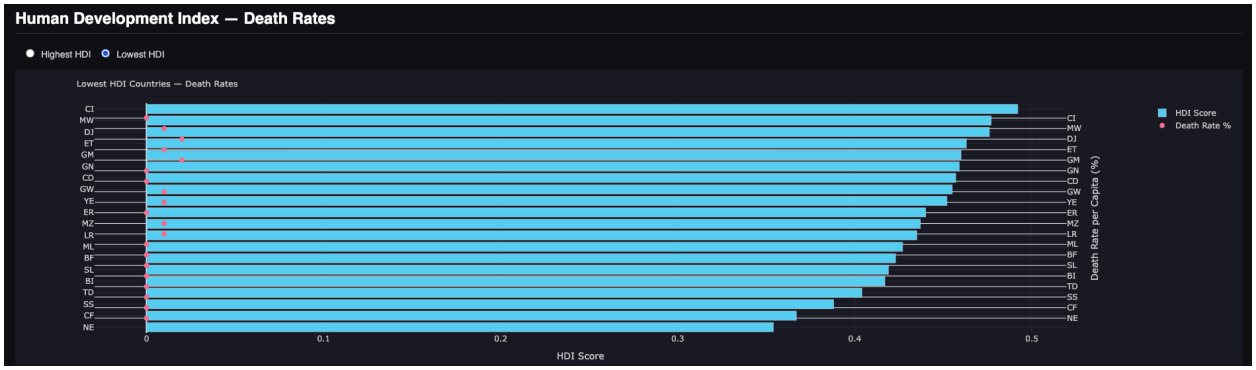


Human Development Index — COVID Cases



Human Development Index — Death Rates




Human Development Index — Death Rates
Highest HDI
Lowest HDI
Lowest HDI Countries — Death Rates
CI
MW
DJ
ET
GM
GN
CD
GW
YE
ER
MZ
LR
ML
BF
SL
BI
TD
SS
CF
NE
0
0.1
0.2
0.3
0.4
0.5
HDI Score
Death Rate per Capita (%)
HDI Score
Death Rate %